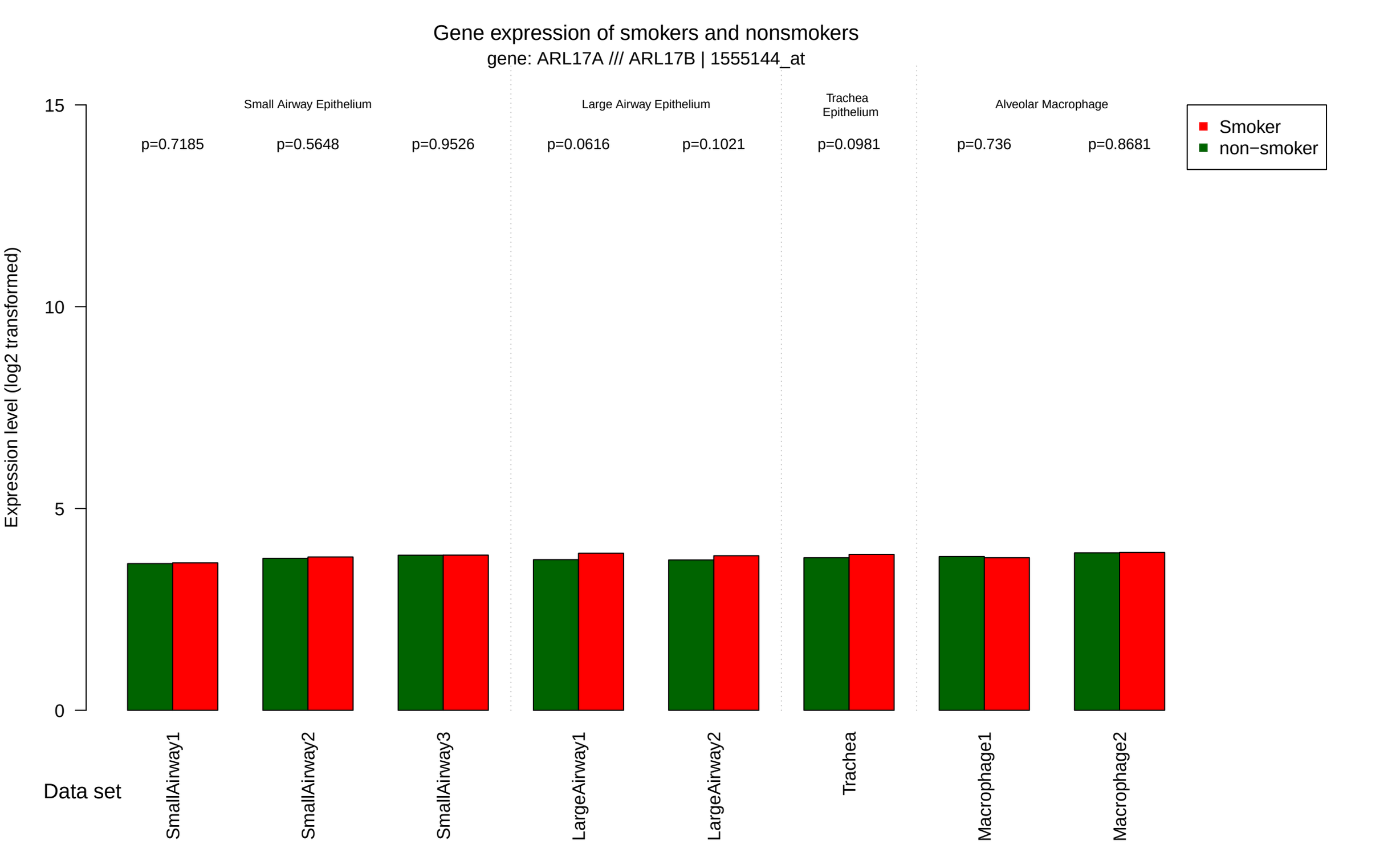
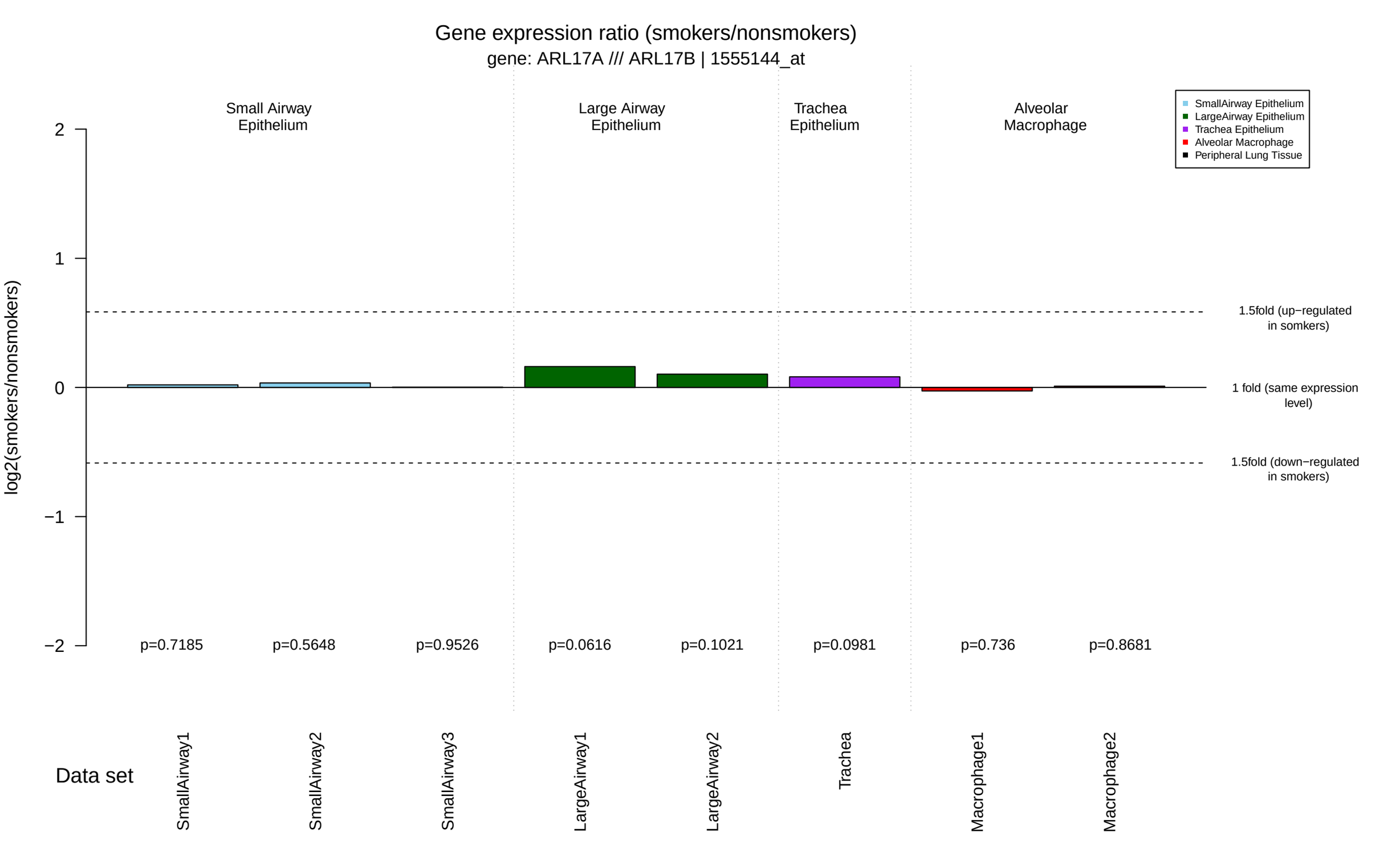
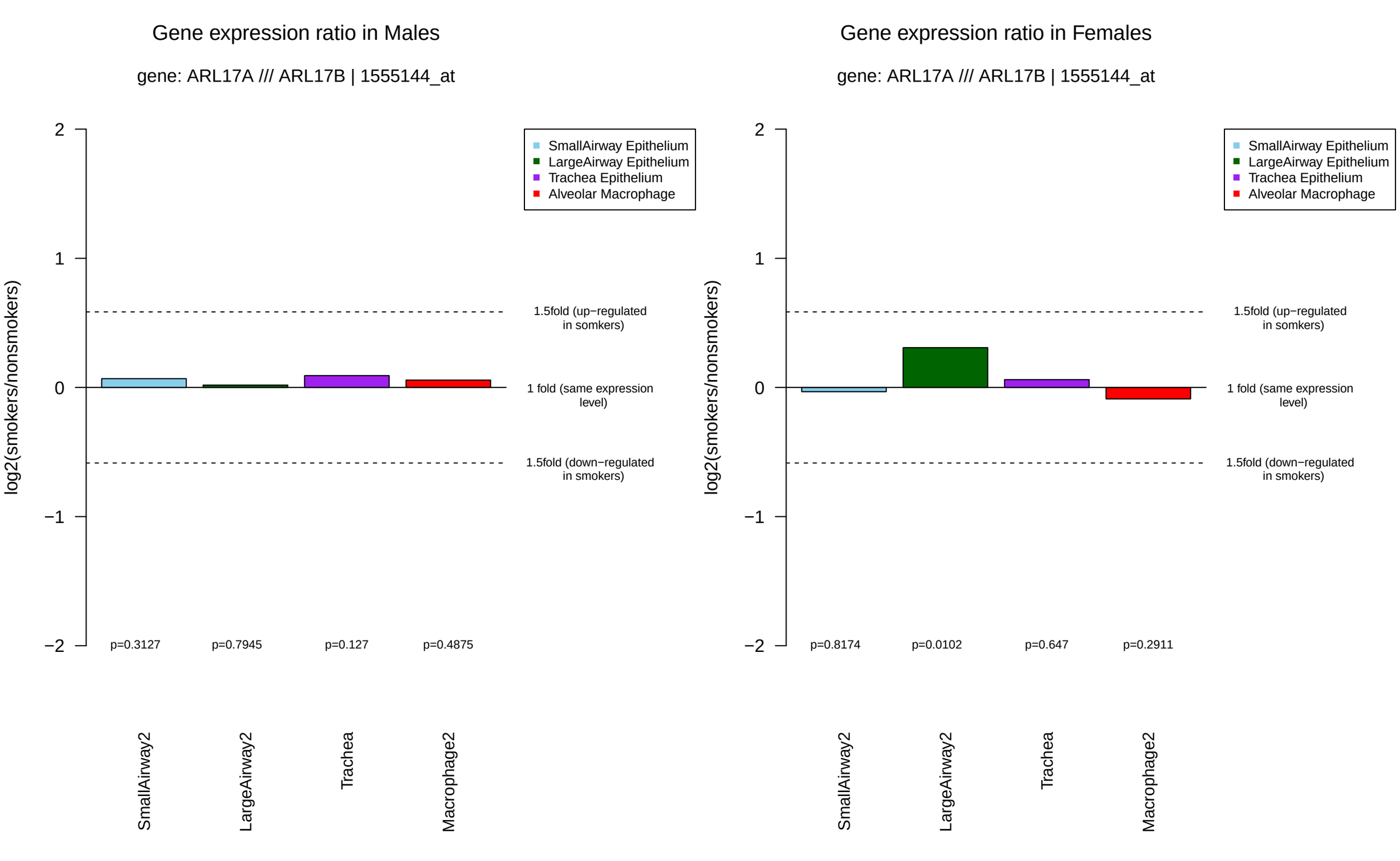
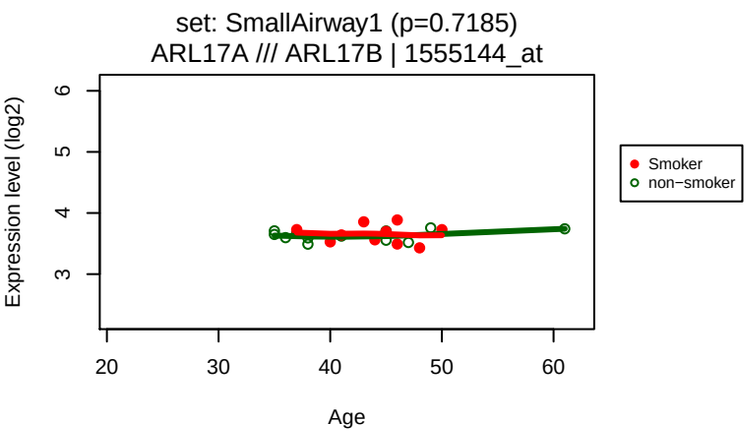
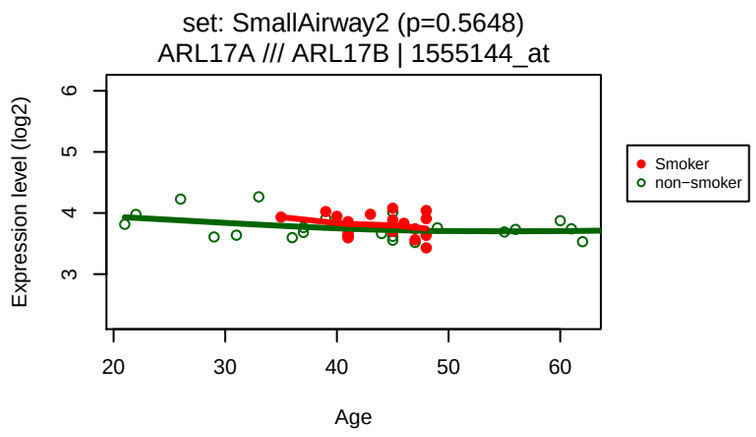
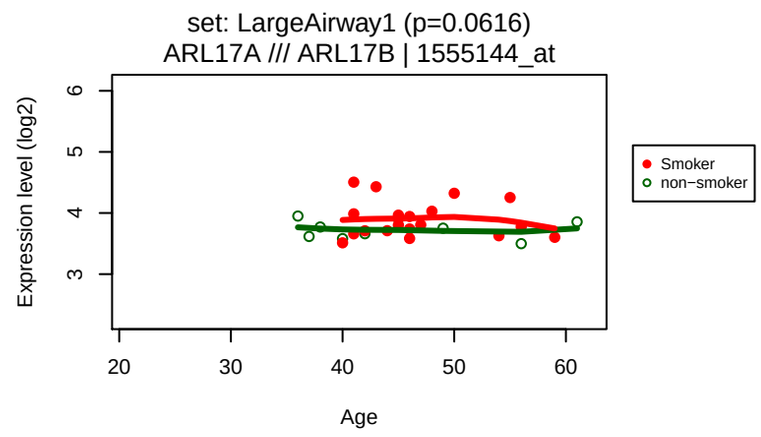

Gene expression of smokers and nonsmokers
gene: ARL17A /// ARL17B | 1555144_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7185
p=0.5648
p=0.9526
p=0.0616
p=0.1021
p=0.0981
p=0.736
p=0.8681
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

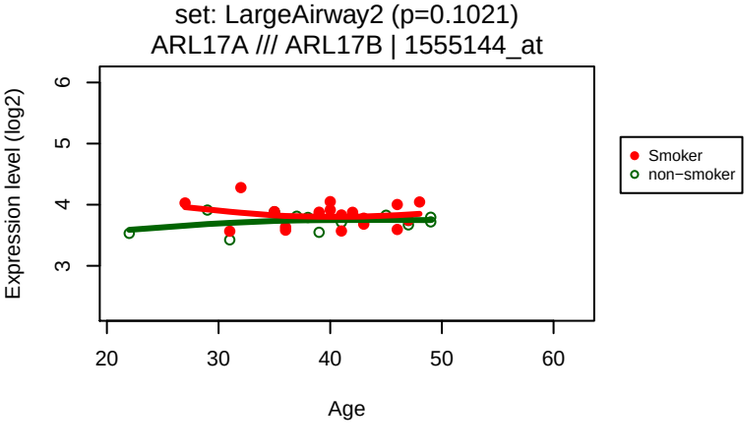

Gene expression ratio (smokers/nonsmokers)
gene: ARL17A /// ARL17B | 1555144_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7185
p=0.5648
p=0.9526
p=0.0616
p=0.1021
p=0.0981
p=0.736
p=0.8681
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

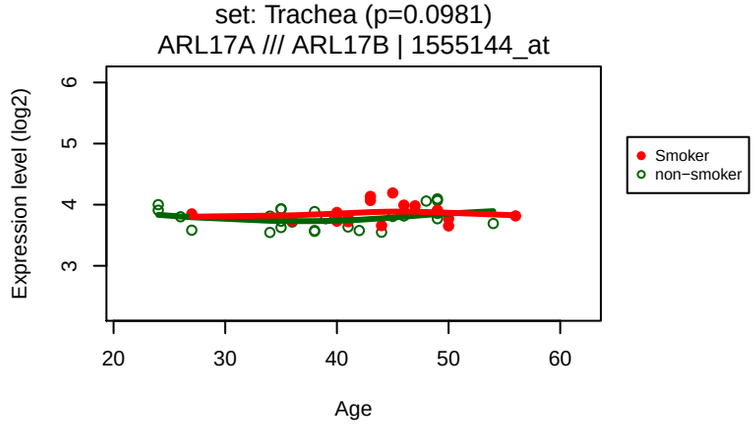

Gene expression ratio in Males
gene: ARL17A /// ARL17B | 1555144_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3127
p=0.7945
p=0.127
p=0.4875
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL17A /// ARL17B | 1555144_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8174
p=0.0102
p=0.647
p=0.2911
SmallAirway2
LargeAirway2
Trachea
Macrophage2

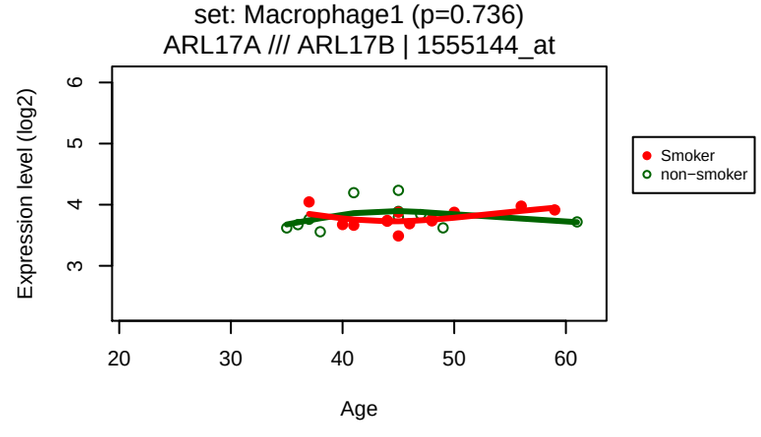

set: SmallAirway1 (p=0.7185)
ARL17A /// ARL17B | 1555144_at
Expression level (log2)
Age
Smoker
non−smoker

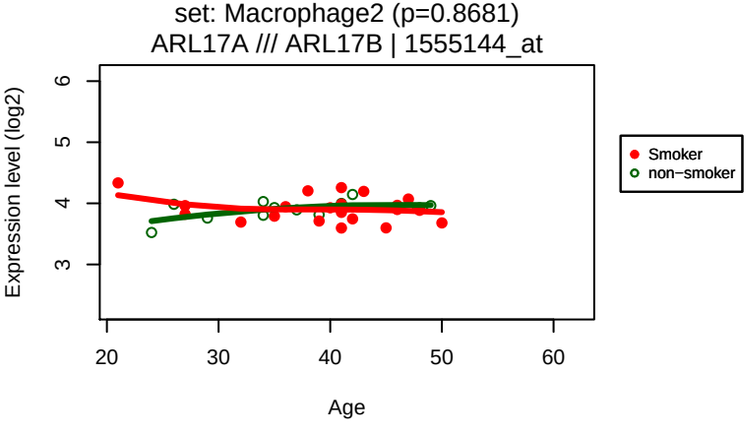

set: SmallAirway2 (p=0.5648)
ARL17A /// ARL17B | 1555144_at
Expression level (log2)
Age
Smoker
non−smoker

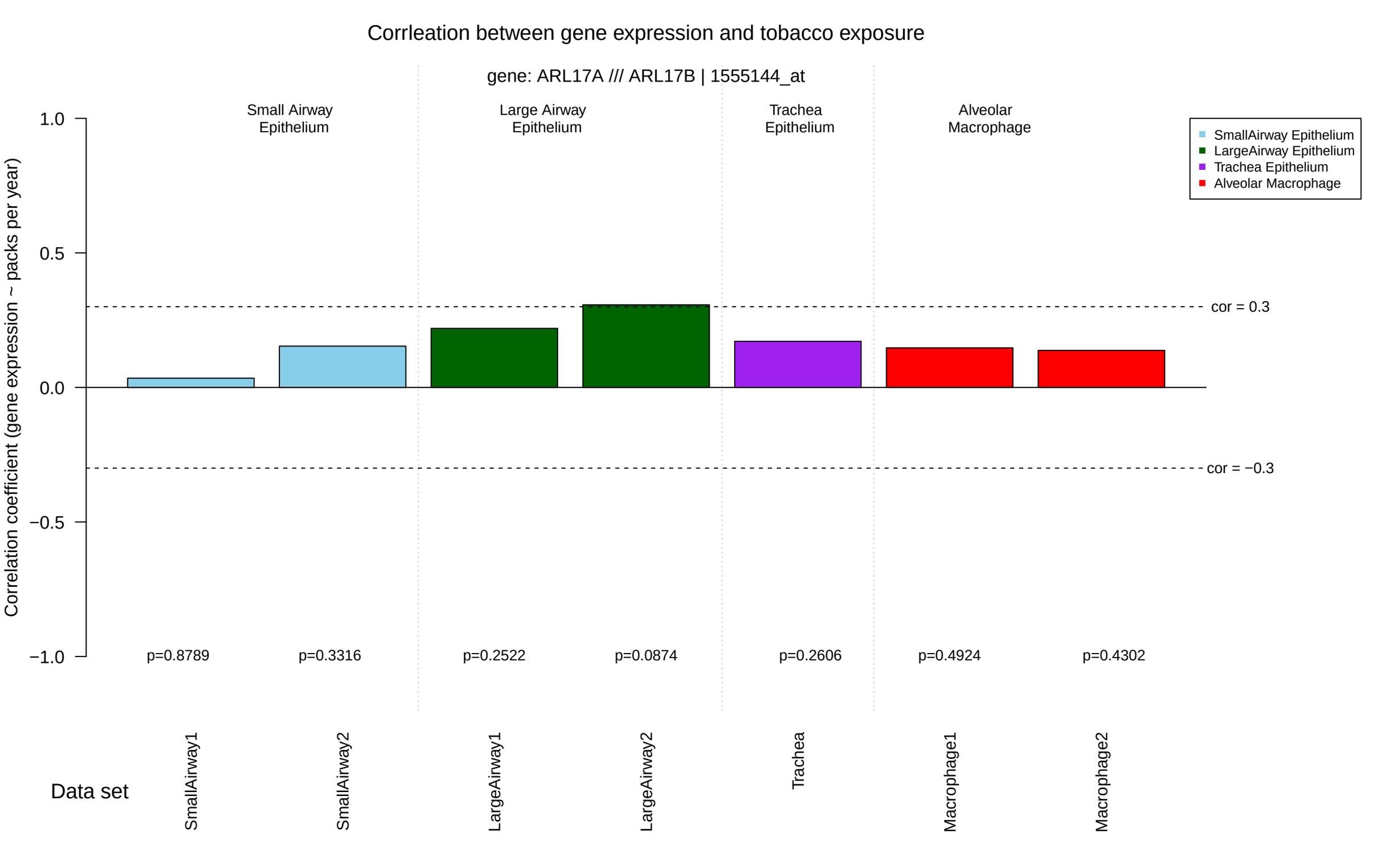

set: LargeAirway1 (p=0.0616)
ARL17A /// ARL17B | 1555144_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1021)
ARL17A /// ARL17B | 1555144_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0981)
ARL17A /// ARL17B | 1555144_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.736)
ARL17A /// ARL17B | 1555144_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8681)
ARL17A /// ARL17B | 1555144_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL17A /// ARL17B | 1555144_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8789
p=0.3316
p=0.2522
p=0.0874
p=0.2606
p=0.4924
p=0.4302
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2